Uniqueness of the Poisson–Dirichlet Invariant Law for the Uniform Split–Merge Chain

Abstract

For the split–merge chain on ranked mass partitions, with uniform fragmentation rate 1 and coagulation acceptance probability $1/\theta$, we prove that for every $\theta > 1$ the classical one-parameter Poisson–Dirichlet law $\mathrm{PD}(\theta)$ is reversible and is the unique invariant Borel probability measure. Reversibility follows from the Gamma–Poisson representation and the Mecke formula. For uniqueness, a stationary threshold-count identity controls the size-biased small-mass tail. This gives a logarithmic bound on the number of cycles in finite central permutation lifts and hence a sublinear relative-entropy bound. An exact cut-and-join intertwining, together with strict binary entropy contraction when $\theta > 1$, then forces every finite lift to be Ewens.

1 Statement and conventions

Let

$$\Omega_1 = \left\{ p = (p_1, p_2, \ldots) : p_1 \geq p_2 \geq \cdots \geq 0, \quad \sum_{i \geq 1} p_i = 1 \right\},$$

equipped with the Borel sigma-field induced by the ℓ^1 topology. Thus all mass is carried by the displayed atoms; dust is excluded.

Fix $\theta > 1$. Given $p \in \Omega_1$, choose I, J independently, conditionally on p, with

$$\mathbb{P}(I = i \mid p) = \mathbb{P}(J = i \mid p) = p_i,$$

and independently choose $U \sim \mathrm{Unif}(0,1)$. If $I = J = i$, replace p_i by Up_i and $(1-U)p_i$, and rank the resulting sequence. If $I \neq J$, merge p_I, p_J into $p_I + p_J$ with probability $1/\theta$, and otherwise leave the state unchanged. Denote this Markov kernel by K_θ.

Let $V_1, V_2, \ldots$ be independent $\mathrm{Beta}(1, \theta)$ variables and set

$$X_1 = V_1, \qquad X_i = V_i \prod_{j<i} (1 - V_j), \quad i \geq 2.$$

The law of the decreasing rearrangement of $(X_i)_{i \geq 1}$ is the classical one-parameter Poisson–Dirichlet law $\mathrm{PD}(\theta)$.

Theorem 1. *For every $\theta > 1$, the law $\mathrm{PD}(\theta)$ is the unique invariant Borel probability measure of K_θ on Ω_1.*

2 Reversibility of the Poisson–Dirichlet law

We use the normalized Gamma-process representation of PD(θ). This representation and its equivalence with the stick-breaking definition above are standard; see Kingman [1] and Pitman [2, Section 3]. The one- and two-point identities used below are the Mecke formula for Poisson processes; see Last and Penrose [3, Chapter 4].

Let η be a Poisson point process on $(0,\infty)$ with intensity

$$\nu_\theta(\,\mathrm{d}x) = \theta x^{-1}e^{-x}\,\mathrm{d}x,$$

and write

$$T_\eta = \int_0^\infty x\,\eta(\,\mathrm{d}x).$$

Its Laplace transform is

$$\mathbb{E}e^{-sT_\eta} = \exp\left\{-\theta\int_0^\infty (1-e^{-sx})x^{-1}e^{-x}\,\mathrm{d}x\right\} = (1+s)^{-\theta}.$$

Thus $0 < T_\eta < \infty$ almost surely. For a point measure ξ with finite positive total mass, let $R(\xi)$ be the decreasing sequence of its atom sizes divided by their total. Then

$$R(\eta) \sim \mathrm{PD}(\theta).$$

Proposition 2. *The law* PD(θ) *is reversible for* K_θ.

Proof. Let $F \geq 0$ be a Borel function on Ω_1^2. The flux through split transitions is

$$\Phi_S(F) = \mathbb{E}\sum_{x\in\eta}\frac{x^2}{T_\eta^2}\int_0^1 F\left(R(\eta), R(\eta-\delta_x+\delta_{ux}+\delta_{(1-u)x})\right)\,\mathrm{d}u.$$

Apply the one-point Mecke formula, and on its right-hand side write T for the total mass of the residual Poisson process. This gives

$$\Phi_S(F) = \theta\int_0^\infty\int_0^1 xe^{-x}\mathbb{E}\left[\frac{F\left(R(\eta+\delta_x), R(\eta+\delta_{ux}+\delta_{(1-u)x})\right)}{(T+x)^2}\right]\,\mathrm{d}u\,\mathrm{d}x. \tag{2.1}$$

Let A be the accepted-merge subkernel. Its reverse-oriented flux is

$$\Phi_A^{\leftarrow}(F) = \frac{1}{\theta}\mathbb{E}\sum_{(y,z)\in\eta^2_{\neq}}\frac{yz}{T_\eta^2}F\left(R(\eta-\delta_y-\delta_z+\delta_{y+z}), R(\eta)\right),$$

where the sum is over ordered distinct atom occurrences. The two-point Mecke formula yields

$$\Phi_A^{\leftarrow}(F) = \theta\int_0^\infty\int_0^\infty e^{-(y+z)}\mathbb{E}\left[\frac{F\left(R(\eta+\delta_{y+z}), R(\eta+\delta_y+\delta_z)\right)}{(T+y+z)^2}\right]\,\mathrm{d}y\,\mathrm{d}z. \tag{2.2}$$

Indeed, the factor multiplying the expectation is

$$\frac{1}{\theta}\left(\theta y^{-1}e^{-y}\right)\left(\theta z^{-1}e^{-z}\right)yz = \theta e^{-(y+z)}.$$

Under

$$x = y + z, \qquad u = \frac{y}{y+z}, \qquad \mathrm{d}y\ \mathrm{d}z = x\ \mathrm{d}x\ \mathrm{d}u,$$

the right-hand side of (2.2) becomes exactly (2.1). Hence

$$\Phi_S(F) = \Phi_A^{\leftarrow}(F). \tag{2.3}$$

There is no extra factor two: the two-point Mecke sum is ordered, and interchanging y, z corresponds to replacing u by $1 - u$.

The rejected-merge subkernel is

$$H(p,\, \mathrm{d}q) = \left(1 - \frac{1}{\theta}\right)\left(1 - \sum_i p_i^2\right)\delta_p(\,\mathrm{d}q).$$

Its edge measure is diagonal and therefore symmetric. Combining this with (2.3) proves

$$\mathrm{PD}(\theta)(\,\mathrm{d}p)K_\theta(p,\, \mathrm{d}q) = \mathrm{PD}(\theta)(\,\mathrm{d}q)K_\theta(q,\, \mathrm{d}p).$$

The ranking-and-normalization map is Borel: its k-th coordinate can be recovered from the measurable counting functions $\xi((r,\infty))$, $r \in \mathbb{Q}_{>0}$, and the measurable total mass. Thus the preceding applications of Mecke and Tonelli are valid for arbitrary nonnegative Borel F. Since the edge measures are finite, the identity extends to bounded signed tests. Reversibility, and hence invariance, follows. □

3 Stationary small-mass estimates

For the remainder of the proof, let μ be an arbitrary invariant Borel probability measure:

$$\mu K_\theta = \mu.$$

Let $P = (P_i)_{i\ge1} \sim \mu$, and put

$$q_2(P) = \sum_i P_i^2.$$

Lemma 3 (Split–merge rate balance)**.**

$$\mathbb{E}_\mu q_2 = \frac{1}{\theta + 1}.$$

Proof. For $\eta > 0$, set

$$N_\eta(P) = \#\{i : P_i > \eta\}.$$

This is a bounded Borel function. In one transition its increment has absolute value at most one. As $\eta \downarrow 0$, that increment converges almost surely to $+1$ on a split, to -1 on an accepted merge, and to 0 on a rejected merge. A size-biased sample selects a positive part almost surely. Stationarity and dominated convergence therefore give

$$0 = \mathbb{P}(\text{split}) - \mathbb{P}(\text{accepted merge}) = \mathbb{E}q_2 - \frac{1}{\theta}\mathbb{E}(1 - q_2).$$

Solving this equation proves the claim. □

For $\varepsilon > 0$, define

$$r_\varepsilon = \sum_{P_i \le \varepsilon} P_i, \qquad t_\varepsilon = \sum_{P_i \le \varepsilon} P_i^2,$$

and

$$A_\varepsilon = \sum_{\substack{i \ne j,\ P_i, P_j \le \varepsilon \\ P_i + P_j > \varepsilon}} P_i P_j.$$

Lemma 4 (Linear small-mass bound)**.** *Whenever* $(\theta+1)\varepsilon < 1$,

$$\mathbb{E} r_\varepsilon \le \frac{2\theta\varepsilon}{1-(\theta+1)\varepsilon}.$$

Proof. We compute the drift of the bounded threshold count

$$N_\varepsilon(P) = \#\{i : P_i > \varepsilon\}.$$

If a selected part has size $x > \varepsilon$, then a uniform split changes the expected count by

$$\mathbb{E}_U \left[\mathbf{1}_{\{Ux > \varepsilon\}} + \mathbf{1}_{\{(1-U)x > \varepsilon\}} - 1 \right] = 1 - \frac{2\varepsilon}{x}.$$

If $x \le \varepsilon$, the change is zero. Hence the split part of the drift is

$$q_2 - t_\varepsilon - 2\varepsilon(1 - r_\varepsilon). \tag{3.1}$$

An accepted merge decreases the count exactly when both selected parts exceed ε. It increases the count exactly when both are at most ε and their sum exceeds ε. All other accepted merges leave the count unchanged. Because the selection is ordered, the merge part of the drift is

$$\frac{1}{\theta} \left[-(1-r_\varepsilon)^2 + q_2 - t_\varepsilon + A_\varepsilon \right]. \tag{3.2}$$

Stationarity of N_ε, together with Lemma 3, gives after rearrangement

$$2\mathbb{E} r_\varepsilon = 2\theta\varepsilon(1 - \mathbb{E} r_\varepsilon) + (\theta+1)\mathbb{E} t_\varepsilon + \mathbb{E} r_\varepsilon^2 - \mathbb{E} A_\varepsilon. \tag{3.3}$$

Now

$$A_\varepsilon \ge 0, \qquad t_\varepsilon \le \varepsilon r_\varepsilon, \qquad r_\varepsilon^2 \le r_\varepsilon.$$

Using these bounds on the right-hand side of (3.3) yields

$$2\mathbb{E} r_\varepsilon \le 2\theta\varepsilon + (\theta+1)\varepsilon \mathbb{E} r_\varepsilon + \mathbb{E} r_\varepsilon.$$

The asserted estimate follows. □

Let X be an annealed size-biased part, meaning

$$\mathbb{E} f(X) = \mathbb{E}_\mu \sum_i P_i f(P_i)$$

for every nonnegative Borel f. Then

$$F(x) := \mathbb{P}(X \le x) = \mathbb{E} r_x.$$

Set

$$x_0 = \frac{1}{2(\theta+1)}.$$

Lemma 4 gives

$$F(x) \le 4\theta x, \qquad 0 \le x \le x_0. \tag{3.4}$$

Lemma 5 (Unseen mass and occupied blocks)**.** *Let m iid paintbox labels be sampled from P. Their expected unseen mass u_m satisfies*

$$u_m = O_\theta(m^{-1}).$$

Consequently, if C_n is the number of occupied atoms among n samples, then

$$\mathbb{E}C_n = O_\theta(\log n).$$

Proof. The expected unseen mass is

$$u_m = \mathbb{E}\sum_i P_i(1-P_i)^m = \mathbb{E}(1-X)^m.$$

For $m \geq 1$, split its Stieltjes integral at x_0. Integration by parts and (3.4) give

$$\begin{aligned} u_m &\leq 2(1-x_0)^m + m\int_0^{x_0}(1-x)^{m-1}F(x)\,\mathrm{d}x \\ &\leq 2(1-x_0)^m + 4\theta m\int_0^1 x(1-x)^{m-1}\,\mathrm{d}x \\ &= 2(1-x_0)^m + \frac{4\theta}{m+1}. \end{aligned} \tag{3.5}$$

Thus $u_m = O_\theta(m^{-1})$, while $u_0 = 1$. The $(m+1)$-st sample creates a new occupied atom precisely when its atom was unseen among the first m samples. Therefore

$$\mathbb{E}C_n = \sum_{m=0}^{n-1} u_m = O_\theta(\log n).$$

□

4 Central permutation lifts and entropy

For a probability law λ on Ω_1, define its central permutation lift $\Phi_n(\lambda)$ as follows. Conditional on a mass partition p, assign labels $1,\ldots,n$ independently to atom i with probability p_i, and place an independent uniform cycle on each nonempty label class. Let

$$\nu_n = \Phi_n(\mu).$$

The family (ν_n) is coherent under deletion and central, meaning invariant under conjugation. Its number of cycles is the number C_n of occupied paintbox atoms.

Let ϵ_n be Ewens measure on S_n:

$$\epsilon_n(\sigma) = \frac{\theta^{c(\sigma)}}{(\theta)_n}, \qquad (\theta)_n = \theta(\theta+1)\cdots(\theta+n-1) = \frac{\Gamma(n+\theta)}{\Gamma(\theta)},$$

where $c(\sigma)$ is the number of cycles of σ.

Lemma 6 (Sublinear finite-level entropy)**.** *For*

$$H_n = D(\nu_n \| \epsilon_n),$$

one has

$$H_n = O_\theta((\log n)^2) = o(n).$$

Proof. If a cycle type is $\lambda = (1^{a_1} 2^{a_2} \cdots)$, then its conjugacy class has cardinality

$$\frac{n!}{\prod_s s^{a_s} a_s!}.$$

Centrality implies that, conditional on the cycle type, ν_n is uniform on this class. Consequently its Shannon entropy satisfies

$$\mathrm{Ent}(\nu_n) \geq \log n! - \mathbb{E}\left[\sum_s a_s \log s + \sum_s \log(a_s!)\right]. \tag{4.1}$$

Each of the two sums in brackets is bounded by $C_n \log n$. Since $\theta > 1$,

$$\begin{aligned} H_n &= \log(\theta)_n - \mathbb{E}C_n \log\theta - \mathrm{Ent}(\nu_n) \\ &\leq \log \frac{(\theta)_n}{n!} + 2\mathbb{E}C_n \log n. \end{aligned} \tag{4.2}$$

By the Gamma-ratio form of Stirling's formula,

$$\log \frac{(\theta)_n}{n!} = O_\theta(\log n).$$

Combining this with Lemma 5 proves the result. □

5 The exact finite cut-and-join representation

Fix $N \geq 2$ and distinguish two driver labels a, b. Let D_N delete them from their cycles. Define a Markov kernel $Q_N^{a,b}$ on S_N by proposing the right multiplication $\sigma \mapsto \sigma(a\,b)$, according to the following rule:

- if a, b lie in the same cycle, transpose them with probability 1;
- if a, b lie in distinct cycles, transpose them with probability $1/\theta$, and otherwise hold.

Right multiplication by $(a\,b)$ cuts one cycle into two or joins two cycles into one.

Lemma 7 (Cut-and-join intertwining)**.** *For every probability law λ on Ω_1,*

$$\Phi_N(\lambda) Q_N^{a,b} D_N = \Phi_{N-2}(\lambda K_\theta). \tag{5.1}$$

Proof. The paintbox marks of a, b are two independent size-biased choices from the mass partition.

Suppose first that their marks agree. Condition on the m retained labels with that mark. A uniform cycle can be written uniquely in the anchored form

$$(a, A, b, B),$$

where A, B are ordered lists partitioning the m retained labels. For a prescribed k-element subset to be the set underlying A, the probability is

$$\frac{k!(m-k)!}{(m+1)!} = \int_0^1 u^k (1-u)^{m-k}\,\mathrm{d}u. \tag{5.2}$$

After transposition and deletion of the drivers, the two arcs become two cycles. Conditional on their vertex sets, their cyclic orders are independent and uniform. Formula (5.2) is precisely the beta-binomial law obtained by first choosing $U \sim \mathrm{Unif}(0,1)$, splitting the selected mass into U and $1-U$ proportions, and then sampling the retained labels. Empty arcs are included.

Suppose next that the driver marks are distinct and the transposition is accepted. The two old cycles are joined. Conditional on two fixed old survivor subsets, the resulting concatenated cycle need not be uniform. However, after averaging over the retained labels' iid latent allocation between the two old atoms, the construction is invariant under every relabeling of their union. It is supported on the set of one-cycle permutations of that union, and the relabeling action is transitive on that set. Hence the resulting cycle is uniform, independently of the unaffected uniform cycles. This is exactly the central lift after merging the two masses. Rejection gives the holding event. These observations prove (5.1), including cases in which an arc or an old driver cycle has no retained label. □

For $a = N-1, b = N$, abbreviate $Q_N = Q_N^{a,b}$. Since μ is invariant and the lift is coherent, Lemma 7 gives

$$\nu_N Q_N D_N = \nu_{N-2} = \nu_N D_N. \tag{5.3}$$

The Ewens law ϵ_N is coherent and Q_N-reversible. Indeed, if s is the state in which the drivers lie in one cycle and $d = s(a\,b)$ is the state in which they lie in two cycles, then

$$\epsilon_N(d) = \theta \epsilon_N(s), \qquad Q_N(s,d) = 1, \qquad Q_N(d,s) = \frac{1}{\theta}. \tag{5.4}$$

Define

$$\delta_N = H_N - D(\nu_N Q_N \| \epsilon_N).$$

Data processing under Q_N, followed by data processing under D_N and the identities (5.3), gives

$$0 \le \delta_N \le H_N - H_{N-2}. \tag{5.5}$$

Set $H_0 = 0$. Summing (5.5) over even N and using Lemma 6 shows that there is a sequence of even integers $N_j \to \infty$ such that

$$\delta_{N_j} \longrightarrow 0. \tag{5.6}$$

6 Binary entropy coercivity and projection

The driver transposition partitions S_N into two-state orbits. Within such an orbit, retain the notation s, d from (5.4). Conditional Ewens weights are

$$\pi(s) = \frac{1}{\theta+1}, \qquad \pi(d) = \frac{\theta}{\theta+1}. \tag{6.1}$$

If an arbitrary conditional input puts probability r on s, one Q_N-step puts probability

$$T(r) = \frac{1-r}{\theta} \tag{6.2}$$

on s. Let

$$h(r) = D\left(\mathrm{Bern}(r) \,\|\, \mathrm{Bern}\left(\frac{1}{\theta+1}\right)\right).$$

Lemma 8 (Uniform binary coercivity)**.** *There exists $c_\theta > 0$, depending only on θ, such that*

$$h(r) - h(T(r)) \geq c_\theta \big(r - T(r)\big)^2, \qquad 0 \leq r \leq 1. \tag{6.3}$$

Consequently,

$$\delta_N \geq c_\theta \|\nu_N - \nu_N Q_N\|_{\mathrm{TV}}^2. \tag{6.4}$$

Proof. Data processing gives $h(r) \geq h(T(r))$. Equality in the log-sum inequality can hold only at

$$r_* = \frac{1}{\theta + 1}.$$

Indeed, because $\theta > 1$, the output state d receives positive reference transition mass from both input states, so equality forces their likelihood ratios to agree. The quotient

$$\frac{h(r) - h(T(r))}{(r - T(r))^2}$$

is continuous away from r_*, strictly positive there, and has the positive limiting value

$$\frac{\theta^2 - 1}{2\theta}$$

at r_*, by Taylor expansion. Compactness of $[0, 1]$ proves (6.3).

For (6.4), decompose relative entropy first by transposition orbit and then within each orbit. The orbit masses are unchanged by Q_N. If w_o is the ν_N-mass of orbit o and r_o its conditional s-mass, then

$$\delta_N = \sum_o w_o (h(r_o) - h(T(r_o))) \geq c_\theta \sum_o w_o (r_o - T(r_o))^2.$$

On the other hand,

$$\|\nu_N - \nu_N Q_N\|_{\mathrm{TV}} = \sum_o w_o |r_o - T(r_o)|.$$

Cauchy–Schwarz, using $\sum_o w_o = 1$, gives (6.4). Orbits of zero ν_N-mass contribute zero, so no positivity assumption on ν_N is needed. □

By (5.6) and Lemma 8,

$$\|\nu_{N_j} - \nu_{N_j} Q_{N_j}\|_{\mathrm{TV}} \longrightarrow 0. \tag{6.5}$$

Centrality permits conjugating the driver pair to $(1, 2)$ without changing this distance.

Fix $k \geq 2$. Let $D_{N,k}$ delete labels $k + 1, \ldots, N$, retaining $1, 2$. Deletion preserves whether $1, 2$ lie in the same cycle. Contracting the two cycle arcs before or after cutting or joining them at $1, 2$ gives the pathwise commutation identity

$$Q_N^{1,2} D_{N,k} = D_{N,k} Q_k^{1,2}. \tag{6.6}$$

Coherence, (6.5), (6.6), and contraction of total variation under a Markov kernel imply

$$\begin{aligned} \|\nu_k - \nu_k Q_k^{1,2}\|_{\mathrm{TV}} &\leq \|\nu_{N_j} - \nu_{N_j} Q_{N_j}^{1,2}\|_{\mathrm{TV}} \\ &\longrightarrow 0. \end{aligned}$$

Therefore

$$\nu_k Q_k^{1,2} = \nu_k. \tag{6.7}$$

Centrality conjugates (6.7) to every transposition pair.

7 Identification of the invariant law

Within a transposition orbit s, d, invariance under $Q_k^{a,b}$ gives

$$\nu_k(d) = \theta \nu_k(s), \tag{7.1}$$

where d has one more cycle than s. Transpositions connect the Cayley graph of S_k, so (7.1) implies

$$\nu_k(\sigma) = C_k^\circ \theta^{c(\sigma)}$$

for a normalizing constant C_k°. The identity

$$\sum_{\sigma \in S_k} \theta^{c(\sigma)} = (\theta)_k \tag{7.2}$$

follows inductively: from a permutation of $k-1$ labels, either make k a singleton, contributing a factor θ, or insert k after one of the $k-1$ existing labels without changing the number of cycles. Thus

$$\nu_k(\sigma) = \frac{\theta^{c(\sigma)}}{(\theta)_k} = \epsilon_k(\sigma) \qquad \text{for every } k. \tag{7.3}$$

After forgetting cyclic orders, (7.3) says that the paintbox partition induced by μ has the Ewens finite-dimensional distributions. These determine the infinite exchangeable partition law. Conditional on any $p \in \Omega_1$, the strong law of large numbers gives, simultaneously for all positive atoms, that each paintbox block has asymptotic frequency equal to the corresponding atom mass. Because there is no dust, the decreasing sequence of positive block frequencies is exactly p. Hence the infinite paintbox partition determines the law μ.

The Ewens exchangeable partition has ranked block frequencies $\mathrm{PD}(\theta)$, by the Gamma-process/GEM representation [1, 2]. Therefore (7.3) implies

$$\mu = \mathrm{PD}(\theta).$$

Together with Proposition 2, this proves Theorem 1.

References

[1] J. F. C. Kingman, Random discrete distributions, *Journal of the Royal Statistical Society. Series B (Methodological)* **37** (1975), 1–22.

[2] J. Pitman, *Combinatorial Stochastic Processes*, Lecture Notes in Mathematics, vol. 1875, Springer, Berlin, 2006.

[3] G. Last and M. Penrose, *Lectures on the Poisson Process*, Institute of Mathematical Statistics Textbooks, vol. 7, Cambridge University Press, Cambridge, 2017.